Supply Chain Sales & Delivery Performance Analysis

1. Project Overview

The Supply Chain Sales & Delivery Performance Analysis project was developed as an end-to-end Data Analyst portfolio project using SQL Server, Power Query, Power BI, and DAX.

The objective was to analyze sales and profitability, compare performance across markets and product categories, evaluate shipping performance, and present the results through an interactive two-page Power BI report.

The project followed the workflow:

Raw CSV > SQL Profiling and Validation > Power Query Cleaning > DAX Measures > Power BI Dashboard > Validation > Business Insights

2. Tools Used

- SQL Server
- SQL Server Management Studio (SSMS)
- Power BI Desktop
- Power Query
- DAX
- CSV Dataset

3. Dataset Overview

The project uses the DataCo Supply Chain practice dataset. Each row represents an order item; an order can contain multiple items.

The main fields include:

- Order ID and Order Item ID
- Order Date, Shipping Date, Order Year and Order Month
- Market, Order Region and Order Country
- Customer Segment and Category Name
- Sales, Order Profit and Order Quantity
- Actual Shipping Days and Scheduled Shipping Days
- Shipping Mode and Delivery Status
- Late Delivery Risk, Shipping Delay Days and Profit Status

The dataset contains:

- 180,519 order-item records
- 65,752 distinct orders
- 180,519 distinct Order Item IDs
- Order dates from 1 January 2015 to 31 January 2018

The 2018 data covers January only and should not be compared with complete years as a full-year result.

4. SQL Data Profiling

The CSV dataset was loaded into SQL Server for exploration and validation. The Power BI query connects to the dbo.dataco_supply_chain table and loads the reporting table as Supply_Chain.

Profiling and validation focused on:

- Total record count
- Distinct Order IDs and Order Item IDs
- Column names and data types
- Categorical values used in slicers and charts
- Shipping-day calculations and date fields

The validated counts were 180,519 records, 65,752 distinct orders and 180,519 distinct order items. These counts establish the reporting grain and prevent order items from being incorrectly presented as distinct orders.

5. SQL Business Analysis

SQL supported source exploration and count validation before the final Power BI analysis. The business analysis covered:

- Overall sales, profit, profit margin and order counts
- Sales by market, region and product category
- Sales share by customer segment
- Monthly sales and profit across selected years
- Late-delivery rates and shipping performance
- Loss-making order items

The source counts were retained as a validation reference for the Power BI dataset.

6. Data Cleaning Using Power Query

The SQL dataset was imported into Power BI and prepared using Power Query.

Column Selection and Standardization

- Removed fields unnecessary for this report, including customer names, email, password, street address, image links and unused identifiers.
- Renamed source columns to readable business names, such as Actual Shipping Days, Scheduled Shipping Days, Order Date and Order Profit.
- Set shipping-day fields and the late-risk flag to appropriate numeric types.

Calculated Columns

- Shipping Delay Days = Actual Shipping Days - Scheduled Shipping Days.
- Delivery Performance = Late when Late Delivery Risk is 1; otherwise Not Late.
- Profit Status = Loss when Order Profit is below zero; otherwise Profit.
- Created Order Year, Order Month and Order Month Number from Order Date.

Shipping variance can be negative (early) or positive (late). Not Late includes canceled items; Profit includes zero-profit items. All 180,519 records were retained.

7. DAX Measures

DAX measures were created to calculate the report KPIs and support interactive analysis.

The main measures include:

- Total Orders, Total Quantity, Total Sales and Total Profit
- Profit Margin
- Late Delivery Records, Late Delivery Rate and Late Delivery Count
- Avg Actual Shipping Days and Avg Scheduled Shipping Days
- Avg Shipping Delay
- Loss Records

The main KPI values for the complete dataset are:

- Total Orders: 65,752
- Total Sales: $36,784,735.01 (approximately $36.78M)
- Total Profit: $3,966,902.97 (approximately $3.97M)
- Profit Margin: 10.78%
- Late Delivery Rate: 54.83% of order items
- Late Delivery Records: 98,977
- Average Actual Shipping Days: 3.50
- Average Scheduled Shipping Days: 2.93
- Average Shipping Variance: 0.57 days
- Loss Records: 33,784 order items

Total Orders uses a distinct count of Order ID. Profit Margin divides total profit by total sales. Late Delivery Rate averages the 0/1 late-risk flag. Shipping averages and loss counts are calculated at order-item level.

8. Power BI Dashboard

A two-page report was developed with a consistent navy-and-teal theme. Red and amber highlight late-delivery, shipping-variance and loss-related metrics.

Page 1: Supply Chain Overview

KPI cards show Total Orders, Total Sales, Total Profit, Profit Margin and Late Delivery Rate.

The analytical visuals include:

- Sales and profit by month, combining the selected years
- Sales share by customer segment
- Top 10 product categories by sales
- Top 10 order regions by sales
- Sales by market

Slicers: Order Year, Market, Customer Segment and Order Region.

8. Power BI Dashboard (continued)

Page 2: Delivery Performance

KPI cards show Late Delivery Rate, average actual shipping days, average scheduled shipping days, average shipping variance and loss-making order items.

The analytical visuals include:

- Order items by delivery status
- Late item rate by shipping mode
- Actual versus scheduled shipping days
- Average shipping variance by market
- Late item rate by market
- Profit by late-risk classification

Slicers: Order Year, Market, Shipping Mode and Delivery Status.

Interactive Features

- Page-navigation buttons and Clear slicers controls
- Synced Order Year and Market selections across both pages
- Chart selections that filter other analytical visuals
- Hover tooltips with related performance measures
- Supply-chain illustrations below the sidebar controls

The monthly chart combines the same month across the selected years. Select one Order Year to examine that year separately. Clearing slicers does not necessarily clear an active chart selection; select the chart item again to remove that selection.

9. SQL Stored Procedures

Stored procedure development is not included in this documented Supply Chain version. SQL work focused on source exploration and validation, while interactive filtering is implemented in Power BI.

10. Data Validation

Validation checked the reporting grain, calculated fields, KPI definitions and preservation of the source data during dashboard development.

The main checks included:

- SQL record-count checks: 180,519 records, 65,752 distinct orders and 180,519 distinct order items.
- Recalculated sales, profit, margin, late-item rate and shipping averages from the model data.
- Confirmed that Shipping Delay Days represents actual minus scheduled days.
- Checked the date range and the partial-year coverage for 2018.
- Verified that dashboard redesigns preserved the data model and existing DAX measures.
- Checked page navigation references, filter interactions and visual alignment.

The model totals used for validation are recorded in Section 7. Detailed SQL-to-DAX reconciliation scripts are not included in this documentation.

11. Key Business Insights

The analysis identified the following observations:

- Total sales were approximately $36.78 million, with approximately $3.97 million in profit and a 10.78% profit margin.
- Europe generated the highest sales at approximately $10.87 million, followed by LATAM at $10.28 million.
- Fishing was the highest-selling product category at approximately $6.93 million, followed by Cleats and Camping & Hiking.
- Consumer customers contributed approximately 51.91% of total sales.
- Approximately 54.83% of order items were flagged as late.
- First Class had the highest late-item rate at approximately 95.32%; Standard Class had the lowest at approximately 38.07%.
- Actual shipping averaged 3.50 days compared with 2.93 scheduled days, producing a net average variance of 0.57 days.
- The dataset contained 33,784 loss-making order items.

The high late-item rate, especially for First Class shipments, identifies an area for further investigation. These comparisons describe the dataset and do not establish the operational cause of delays. Total sales should be considered alongside order volume, profit and service performance.

12. Project Outcome

This project provided hands-on experience with an end-to-end Data Analyst workflow: SQL profiling and validation > Power Query data preparation > DAX measures > interactive Power BI reporting > business insight generation.

The completed report presents commercial and logistics performance across two consistent pages. It demonstrates practical skills in data preparation, KPI definition, business analysis, dashboard design and validation.

13. Dataset Disclaimer

This project was created for educational and portfolio purposes using a publicly available DataCo Supply Chain practice dataset. It is not affiliated with DataCo or an actual logistics operator, and the findings should not be presented as current operational performance.